

Kiriko — A GPU-Accelerated Sturm–Liouville Spectral Framework for Variable-Density Anelastic Flow

From pseudospectral Navier–Stokes baseline to machine-precision Lane–Emden g-mode closure, with an externally validated Chebyshev eigenproblem and a proven full-Galerkin time-domain operator

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Abstract

We present **Kiriko**, a GPU-accelerated spectral framework for the linear anelastic equations on a stratified background, and document the sequence of derivations, numerical experiments, and operator-level diagnostics that were required to bring it from a uniform-density pseudospectral baseline to a fully closed-loop time-domain solver at machine precision. The core algorithmic contribution is a seven-step Sturm–Liouville (SL) Poisson pipeline that reduces the variable-coefficient pressure equation $\nabla \cdot (\rho_0^{-1} \nabla p) = f$ to a diagonal problem in the basis of the Liouville-regularised Schrödinger operator, executed on a single GPU through batched cuBLAS ZGEMM and cuFFT. On the Boussinesq limit the spatial solver attains $\|\pi_{\text{num}} - \pi_{\text{exact}}\|_{L^2} \approx 4.6 \times 10^{-14}$, and its eigenstructure is validated component-wise against the stellar-pulsation code GYRE, achieving a maximum relative difference of 3.6×10^{-5} on the first ten $\ell = 1$ g-modes of the Lane–Emden $n = 3$ polytrope. A second set of results characterises the role of the polytropic surface exponent $\sigma = n$ in controlling spectral convergence: integer σ (analytic background density) admits exponential Chebyshev convergence to 6.7×10^{-11} at $N = 64$, while half-integer σ produces a branch-point obstruction limiting convergence to the algebraic rate N^{-2} irrespective of prefactor substitutions. Finally, we identify a latent operator mismatch in the naive time-domain discretisation—the Lane–Emden evolution leaks 5.1×10^{-5} per step irrespective of spatial resolution—and prove, through a full-Galerkin assembled-matrix formulation advanced by explicit Runge–Kutta, that the mismatch collapses to 5.1×10^{-18} , thirteen orders of magnitude below the primitive node-space scheme. The resulting implementation path is transcribed into a concrete specification for porting the assembled operator to CUDA, and extended to a polar-coordinate annulus geometry as a blueprint for future stellar-interior simulations.

1. Introduction

1.1 Motivation

Pseudospectral methods are the method of choice for high-Reynolds-number incompressible turbulence in doubly periodic domains: Fourier bases diagonalise the Laplacian, the pressure projection collapses to point-wise division by $|\mathbf{k}|^2$, and the dominant cost reduces to $O(N \log N)$ fast transforms. The baseline solver

of this project (`pseudo_spectral_solver`) realises this programme for the two-dimensional vorticity–streamfunction form and has been validated to deliver an effective viscosity within 4% of the prescribed value at $\text{Re} = 10^6$, $N^2 = 1024^2$, $t = 40$.

Stellar interiors, however, are intrinsically stratified: the radial density background $\rho_0(r)$ varies by many orders of magnitude between the core and the surface, and buoyancy supplies the dominant restoring force for the entire family of gravity modes (g-modes). Pure Fourier pseudospectra are useless on such domains: the pressure-Poisson operator becomes variable-coefficient,

$$\nabla \cdot (\rho_0^{-1}(y) \nabla p) = f(\mathbf{x}), \quad (1.1)$$

so that Fourier modes are no longer eigenfunctions, and the cost balloons to $O(N^3)$ unless the vertical direction is treated by a basis adapted to ρ_0 . The anelastic approximation removes acoustic waves but retains stratification, which is precisely the regime that the uniform-density pseudospectral infrastructure cannot address.

1.2 Contribution

This paper documents **Kiriko**, the project-wide effort to extend the pseudospectral baseline to variable-density stratified flow on GPUs. Our contributions divide into four blocks:

1. **Algorithm (Section 3).** A Liouville-regularised Sturm–Liouville expansion for the stratified pressure Poisson equation, implemented as a seven-step GPU pipeline.
2. **Validation (Section 4).** Component-wise verification of the Chebyshev eigenproblem against GYRE, reaching a worst-case relative difference of 3.6×10^{-5} on the first ten $\ell = 1$ g-modes.
3. **Spectral convergence dichotomy (Section 5).** A previously undocumented integer-versus-fractional cliff in the polytropic index n , together with a rigorous approximation-theoretic explanation.
4. **Operator-mismatch diagnosis and closure (Section 6).** A negative result—namely that naive node-space time integration leaks 5.1×10^{-5} per step on Lane–Emden $n = 3/2$ regardless of grid refinement—and a proven full-Galerkin remedy that restores machine precision through assembled L and R matrices.

We close (Section 7) with the design of a polar-coordinate extension that transcribes the Cartesian framework to an annular stellar-interior geometry.

2. Background and related work

The variable-coefficient pressure Poisson equation (1.1) admits three families of solvers: algebraic multi-grid, domain-decomposition Schur complements, and spectrally expanded bases. Only the third preserves the $O(N \log N)$ scaling that makes pseudospectral fluid solvers competitive, and only when the vertical basis can be chosen so that the variable coefficient becomes a multiplier on the expansion coefficients. The Liouville transformation $p(y) = \sqrt{\rho_0(y)} q(y)$ converts the stratified self-adjoint operator into canonical Schrödinger form with an explicit potential $W(y)$ or its reduced counterpart $\tilde{W}(y)$, whose sign depends on the choice of variables—an issue we analyse in §3.2. The resulting eigenvalue problem is, in the absence of stratification, equivalent to a Fourier problem; in the presence of stratification it introduces a y -dependent

potential whose eigenfunctions are computed once, stored, and used thereafter as a diagonalising basis. Stellar oscillation codes (GYRE, ADIPLS) use related expansions in finite-difference or finite-element form, but none are GPU-resident or embedded in a full nonlinear anelastic time-stepper.

The two-dimensional pseudospectral baseline for this project uses cuFFT R2C/C2R with IFRK3 integrating-factor Runge–Kutta time stepping, Orszag-style skew-symmetric convection, and a circular 2/3 dealiasing rule. We refer to the design note `pseudo_spectral_design_2026-05-01` for numerical details; for the present paper the baseline functions as a validated backdrop against which the stratified extensions are measured.

3. Algorithmic formulation

3.1 Variable-density pressure Poisson

In the anelastic approximation the pressure enforces incompressibility through

$$\nabla \cdot (\rho_0(y) \mathbf{u}) = 0, \quad \nabla \cdot (\rho_0^{-1} \nabla p) = f(\mathbf{x}). \quad (3.1)$$

Fourier expansion in the periodic direction x reduces (3.1) to a one-dimensional variable-coefficient ODE for each horizontal wavenumber k_x :

$$\frac{d}{dy} \left[\frac{1}{\rho_0(y)} \frac{d\hat{p}}{dy} \right] - \frac{k_x^2}{\rho_0(y)} \hat{p} = \hat{f}(k_x, y). \quad (3.2)$$

3.2 Liouville regularisation

The standard substitution $\hat{p} = \sqrt{\rho_0} q$ converts (3.2) into Schrödinger canonical form

$$-q'' + [k_x^2 - W(y)]q = g(y), \quad W(y) = \frac{\rho_0''}{2\rho_0} - \frac{3(\rho_0')^2}{4\rho_0^2}. \quad (3.3)$$

Working instead with the reduced pressure $\pi = p/\rho_0$ and the substitution $\pi = \rho_0^{-1/2} q$ yields an analogous equation with the modified potential

$$\widetilde{W}(y) = \frac{(\rho_0')^2}{4\rho_0^2} - \frac{\rho_0''}{2\rho_0}. \quad (3.4)$$

For Lane–Emden polytropes the two potentials differ in sign: $W(y)$ is attractive near the surface (possibly unbounded), while $\widetilde{W}(y)$ is repulsive and better bounded after a density cutoff. Experiment A (`reduced_pressure_experiments_2026-05-02`) reports a nine-fold reduction in q-norm error for the reduced-pressure form at matched mode count, making (3.4) the operational choice.

3.3 Sturm–Liouville eigenexpansion

Let $A = -D^2 - \text{diag}(\widetilde{W})$ on the interior Chebyshev–Gauss–Lobatto (CGL) nodes. The operator is self-adjoint under the Clenshaw–Curtis weighted inner product $\langle u, v \rangle_{\text{cc}} = \sum w_j u_j v_j$, but is not Euclidean-symmetric on the collocation grid. Diagonalisation yields eigenpairs $\{(\mu_n, \psi_n)\}$ independent of k_x : for

each horizontal wavenumber the pressure equation becomes the diagonal relation

$$Q_n(k_x) = -\frac{G_n(k_x)}{\mu_n + k_x^2}, \quad (3.5)$$

where G_n and Q_n are the projections of the source and solution onto ψ_n .

3.4 Seven-step GPU pipeline

End-to-end inversion of (3.1) on the GPU proceeds by

$$\text{FFT}_x \rightarrow \times \rho_0^{-1/2} \rightarrow \Psi_{\text{fwd}}^\top \rightarrow \text{diag solve} \rightarrow \Psi \rightarrow \times \rho_0^{1/2} \rightarrow \text{IFFT}_x. \quad (3.6)$$

Steps 1 and 7 are cuFFT R2C/C2R; steps 3 and 5 are ZGEMM of a stored $N_y \times N_y$ eigenmatrix against the k_x -strided field; steps 2, 4, 6 are pointwise kernels. Row-major storage of the coefficient array $\hat{G}[j_y, k_x]$ is bit-identical to column-major ($n_{\text{modes}} \times n_h$), so no transpose is required.

4. GPU implementation and manufactured-solution validation

4.1 Eigenstructure on the GPU

The non-Euclidean symmetry of A rules out `cusolverDnDsyevd`. Naively symmetrising $(A + A^\top)/2$ corrupts μ_1 for the Boussinesq limit $\widetilde{W} \equiv 0$ by 5.5%: the symmetrised eigenvalue is 9.329094 against the exact $\pi^2 = 9.869604$. We instead use the non-symmetric solver `cusolverDnXgeev` (CUDA 11.6+) with a real matrix promoted to `CUDA_C_64F`; the imaginary parts of the returned eigenvalues are below 10^{-6} for well-posed problems, and residual eigenvalue agreement with the Python `scipy.linalg.eig` reference is at 10^{-5} rel. Eigenvectors are re-normalised to $\langle \psi_n, \psi_n \rangle_{\text{cc}} = 1$ after the GPU solver returns Euclidean-normalised output.

4.2 Manufactured-solution convergence on the Boussinesq limit

With $\rho_0 \equiv 1$, $\widetilde{W} \equiv 0$, and analytic $\pi_\star(x, y) = \sin(k_x x) \sin(\pi y/L_y)$, the seven-step pipeline returns machine-precision residuals at every resolution tested:

N_y	N_{modes}	$\ \pi_{\text{num}} - \pi_\star\ _{L^2}$
64	32	4.6×10^{-14}
128	64	4.6×10^{-14}
256	128	4.6×10^{-14}
512	256	4.6×10^{-14}

Resolution-independence in the Boussinesq limit is the expected hallmark of a spectrally exact solver whose only error source is floating-point rounding.

4.3 Convergence on Lane–Emden $n = 3/2$

Introducing the half-integer polytrope $\rho_0(y) \sim (L_y - y)^{3/2}$ with a density cutoff $\rho > 10^{-2}$ gives an algebraic rate:

N_y	N_{modes}	L^2 relative error	Reduction factor
64	32	2.88×10^{-4}	—
128	64	5.74×10^{-5}	$5.0\times$
256	128	1.10×10^{-5}	$5.2\times$
512	256	2.04×10^{-6}	$5.4\times$

The observed rate is compatible with $N_y^{-2.3}$, well below the spectral expectation. Section 5 explains why.

4.4 Full-gravity four-variable g-mode eigenproblem vs GYRE

The Chebyshev collocation solver for the full-gravity four-variable linearised stellar pulsation system (`solve_gmode_full_chebyshev`) is validated on the shipped GYRE Lane–Emden $n = 3$ polytrope at $\ell = 1$. Reference frequencies are produced by a live GYRE run with `diff_scheme = COLLOC_GL6` on the 1001-row structure dump `/tmp/gyre_run/poly3.txt`. Two pipeline patches were required for the CUDA binary to ingest the GYRE dump: an explicit `<algorithm> include` for `std::sort`, and a non-finite-row filter in the reader to discard the surface line containing IEEE Infinity in V_2 and A^* ; the latter was silently benign for Python benchmarks whose resampling never approached $x = 1$ but poisoned every row of the dense Q matrix in the CUDA LU factorisation (`cusolverDnDgetrfBatched` returned `info = 256`).

After these fixes, at $N = 96$ (DOF= 388) the CUDA solver reproduces GYRE to machine-tolerable precision:

n_g	ω_{CUDA}^2	ω_{GYRE}^2	rel err
1	2.515930×10^0	2.515928×10^0	8.6×10^{-7}
2	1.285715×10^0	1.285708×10^0	5.4×10^{-6}
3	7.757401×10^{-1}	7.757328×10^{-1}	9.4×10^{-6}
4	5.177826×10^{-1}	5.177760×10^{-1}	1.3×10^{-5}
5	3.699317×10^{-1}	3.699255×10^{-1}	1.7×10^{-5}
6	2.775088×10^{-1}	2.775028×10^{-1}	2.2×10^{-5}
7	2.159323×10^{-1}	2.159266×10^{-1}	2.6×10^{-5}
8	1.728587×10^{-1}	1.728536×10^{-1}	2.9×10^{-5}
9	1.415487×10^{-1}	1.415441×10^{-1}	3.3×10^{-5}
10	1.180727×10^{-1}	1.180684×10^{-1}	3.6×10^{-5}

The maximum relative difference across $n_g = 1, \dots, 10$ is 3.6×10^{-5} ; the monotone growth with n_g is interpolation-dominated (linear interpolation of GYRE’s native 1001-point mesh onto the CGL collocation grid). Parallel Python pipelines pass at the same tolerance: the two-variable Cowling solver agrees with GYRE’s Cowling spectrum at 5.6×10^{-4} , and the four-variable full-gravity Python solver agrees with GYRE at 5.3×10^{-4} , both well inside the 10^{-2} PASS threshold. At a coarser $N = 48$ the CUDA solver exhibits two spurious head modes with $\omega^2 \approx 500, 443$ that correspond to classifier misidentifications in

the low-resolution propagation cavity; identical head contamination is reproduced by the Python verifier, confirming the artefact is classification-rooted rather than a CUDA defect.

A further topological check confirms the Sturm oscillation theorem discretely: all 21 computed eigenfunctions $\psi_0, \dots, \psi_{20}$ possess exactly n interior zeros, demonstrating that the non-Euclidean symmetric operator preserves the ordering of the continuous problem.

5. Polytropic index and spectral convergence

5.1 The integer-versus-fractional cliff

A focused convergence study on the reduced-pressure Poisson problem with a manufactured exact solution $\pi_\star(r) = \sin(2\pi r)$ and $\rho(r) = (1 - r)^\sigma$ on $r \in [0, 1]$ produces a striking bifurcation as a function of the surface exponent σ .

For $\sigma = 3$ (Lane–Emden $n = 3$, the Eddington polytrope for radiation-pressure-supported stars):

N	raw error
16	8.5×10^{-8}
32	$\sim 10^{-10}$
64	6.7×10^{-11}
128	$\sim 10^{-9}$ (round-off climb)
256	3.2×10^{-9}

Raw Chebyshev collocation achieves machine precision with no prefactor, no Jacobi weight, and no coordinate stretch.

For $\sigma = 3/2$ (Lane–Emden $n = 3/2$, the convective-core polytrope):

N	raw error
16	2.7×10^{-3}
64	1.8×10^{-4}
256	1.1×10^{-5}

The convergence rate tracks $N^{-2.0}$ robustly, and **no choice of analytic prefactor** $\pi = t^\alpha u$ **restores the spectral rate**: the experiment swept $\alpha \in \{1/4, -1/2, -3/4\}$ and found only degradation.

5.2 Approximation-theoretic explanation

Chebyshev expansions of an analytic function on $[0, R]$ have exponentially decaying coefficients precisely when the function admits analytic continuation to a neighbourhood of $[0, R]$ in the complex plane (Trefethen, *Approximation Theory and Approximation Practice*, Thm 7.2).

- $\rho(r) = (R - r)^3$ is a *polynomial*; its Chebyshev expansion terminates at four terms, and $\rho\pi$ inherits the smoothness of π .

- $\rho(r) = (R - r)^{3/2}$ is analytic on $[0, R)$ but has a branch-point singularity at $r = R$; its Chebyshev coefficients decay only as $N^{-\sigma-1/2} \sim N^{-2}$.

Since the Lane–Emden boundary behaviour is $\theta(\xi) \sim (\xi_1 - \xi)$ and $\rho = \theta^n$, the polytropic index *is* the surface exponent:

$$\rho(r) \sim (\xi_1 - \xi)^n \sim (R - r)^n.$$

Integer n yields polynomial density; fractional n injects a branch point that the polynomial basis cannot resolve exponentially.

Polytrope	Physical context	n	Chebyshev rate
$n = 1$	non-relativistic degenerate envelope	1	$N^{-3/2}$ algebraic
$n = 1.5$	convective core / fully convective star	1.5	N^{-2} algebraic
$n = 2$	approximate main-sequence envelope	2	$N^{-5/2}$ algebraic
$n = 3$	Eddington, radiation-pressure supported star	3	spectral (exponential)
$n = 3.25$	giant hydrogen envelope	3.25	$N^{-7/2}$ algebraic

The Eddington polytrope—historically ubiquitous because it produces the mass–radius relation for radiation-supported stars—is the unique classical polytropic choice at which raw Chebyshev achieves spectral convergence. To our knowledge this coincidence between physical and numerical regularity has not been made explicit in the prior literature.

5.3 What works and what does not

Three remedies for fractional σ were considered. The Liouville substitution $\pi = \rho^{-1/2}q$ transforms the operator but not the polynomial-expandability of the unknown, and fails. Power prefactors $\pi = t^\alpha u$ with fractional α introduce a branch point of their own and also fail. The remaining options are basis replacement—Jacobi expansions with weight $(1 - r)^\sigma$ as in Dedalus, which absorb the singularity into the basis functions and restore spectral convergence for any $\sigma > -1$ —and coordinate stretching (Kosloff–Tal–Ezer), which concentrates grid points near the surface. Both live outside the Liouville framework and are earmarked for Phase 2+ of the project.

6. Operator mismatch and the full-Galerkin closure

6.1 Symptom

When the SL spatial solver is used as a pressure projection inside the time stepper of `AnelasticSLSolver`, the Boussinesq Kelvin–Helmholtz test and the Boussinesq g-mode benchmark both close to well within spectral tolerance. A long-time run of the boxed Lane–Emden $n = 3/2$ g-mode, however, exhibits a FFT peak error of -9.4% and an eigenmode deviation-per-step of 6×10^{-4} —large enough that the resolved time-domain signal drifts monotonically away from the correct eigenfrequency after a few tens of wave periods.

6.2 Diagnostic hierarchy

Four candidate explanations were systematically eliminated:

Path	Hypothesis	Outcome
A	EVP uses Fourier q -space (density absorbed into variable)	Lane–Emden dev/step 6.0×10^{-4} ; 13% improvement
B	EVP uses SL-basis q -space (Galerkin-consistent with Poisson step)	Lane–Emden dev/step 5.9×10^{-4} ; 4% improvement
C	Time domain recast in momentum variable $\varphi = \rho v$	Toy 6.8×10^{-5} ; rejected as inadequate
D	Time domain advanced by the assembled operator itself	Lane–Emden dev/step 5.1×10^{-18} ; machine precision

Paths A and B were implemented in CUDA, and path C was prototyped in Python prior to the CUDA investment. None of them closes the Lane–Emden loop because each only fixes a single piece (the Poisson step or the density coefficient) of a composed linear operator whose remaining pieces—the buoyancy multiplication by $N^2(y)$, the continuity multiplication by $\rho'(y)/\rho(y)$, and the Chebyshev derivative `apply_dy` in node space—each scatter the eigenvector by a finite amount in the discrete norm.

6.3 The full-Galerkin closure

After eliminating u_x through continuity and b through the buoyancy equation, the linear anelastic system reduces to the second-order ODE

$$L \ddot{V} = -R V, \quad L = -D \operatorname{diag}(\rho) D + k^2 \operatorname{diag}(\rho), \quad R = k^2 \operatorname{diag}(N^2 \rho). \quad (6.1)$$

The eigenproblem $Rv = \omega^2 Lv$ is the *same discretisation* of (6.1) that the time stepper should advance. If, and only if, the time stepper is evaluated as $\ddot{V} = -L^{-1}RV$ through matrix–vector products against the *same* assembled pair (L, R) , the eigenvector of the spatial problem is automatically an eigenvector of the discrete time operator, and a Runge–Kutta substep preserves it exactly up to time-integration error:

$$V(t) = V_{\text{EVP}} \cos(\omega t), \quad W(t) = -\omega V_{\text{EVP}} \sin(\omega t),$$

with dev/step floored at floating-point rounding.

6.4 Numerical proof

A Python driver (`scripts/full_galerkin_closure_test.py`) compares three time integrators on identical initial data: full-Galerkin v -space, full-Galerkin φ -space, and the primitive node-space scheme that mirrors the CUDA implementation. At $N_y = 64$, RK4, $\Delta t = 10^{-4}$, initial amplitude 10^{-8} , 100 steps:

Case	full-Galerkin v	full-Galerkin φ	primitive (CUDA-like)
Boussinesq	3.2×10^{-18}	3.2×10^{-18}	3.3×10^{-18}
Lane–Emden 3/2	5.1×10^{-18}	4.9×10^{-18}	5.1×10^{-5}

A sweep over $N_y \in \{32, 64, 96, 128, 192\}$ produces full-Galerkin dev/step in $[1.8 \times 10^{-18}, 2.6 \times 10^{-17}]$ and a primitive dev/step **constant at** 5.1×10^{-5} , confirming that the defect is an operator design flaw rather than a discretisation accuracy issue: adding grid points cannot cure it.

6.5 Implementation specification

The CUDA path implied by the proof is a three-to-four-day rewrite:

Task	Effort
Allocate <code>d_Linv_R_kx[]</code> of total $n_h \cdot (n_y - 2)^2$ doubles	0.5 d
Assemble L_{k_x}, R_{k_x} on the host using existing D, ρ, N^2	0.5 d
Compute $M_{k_x} = L_{k_x}^{-1} R_{k_x}$ via <code>cusolverDnDgetrs</code> or LAPACK	0.5 d
Replace the linear part of <code>compute_rhs_uv</code> with a per- k_x DGEMM	1.0 d
Eliminate the buoyancy variable or assemble $N^2 \cdot I$ in the same way	0.5 d
Boussinesq regression plus Lane–Emden dev/step $\leq 10^{-10}$ test	1.0 d

Memory cost is trivial: $33 \cdot 62^2 \cdot 8$ bytes ≈ 1 MB for a 64×64 Cartesian grid. FLOP cost per RK3 substep is $n_y^2 \cdot n_h \approx 135$ kFLOP, slightly below the $n_y^2 \cdot n_x \approx 262$ kFLOP of the current `apply_dy` path.

6.6 Methodological observation

The diagnostic sequence deserves comment in its own right. Paths A and B each required roughly one engineer-day of CUDA development and validated at 10^{-4} —negligibly better than the baseline. Path C was prototyped in Python in under an hour (`scripts/path_c_td_benchmark.py`) and rejected for insufficient improvement before any CUDA work was committed. Path D was identified and validated in an additional

hour (`scripts/full_galerkin_closure_test.py`). Two Python prototypes, totalling approximately two hours of engineer time, converted what would have been a seven- to ten-day speculative CUDA rewrite into a three- to four-day implementation of a proven specification.

7. Extension to polar geometry

The Cartesian framework transcribes directly to an axisymmetric 2D annulus $(r, \phi) \in [r_{\text{in}}, r_{\text{out}}] \times [0, 2\pi)$, with the azimuthal coordinate replacing x (Fourier, periodic) and the radial coordinate replacing y (Chebyshev on a finite interval, Dirichlet on the impermeable walls). Reduced-pressure anelastic linearisation eliminates $\hat{U}_\phi$ through continuity and $\hat{b}$ through the buoyancy equation, yielding

$$L_m \ddot{\hat{U}}_r = -R_m \hat{U}_r,$$

with L_m built from the compound operator $(1/r) \partial_r [r \rho_0 \cdot] \cdot (1/\rho_0) \partial_r(\cdot)$ and $R_m = N^2(r) \mathbf{I}$. The $m = 0$ mode corresponds to radial pulsations and is treated separately. The assembled-matrix time stepper of Section 6 transfers unmodified, exchanging k_x for the azimuthal index m ; the seven-step pressure pipeline transfers in the same way. Inner wall $r_{\text{in}} > 0$ sidesteps the coordinate singularity at the origin, reserving basis replacement (Zernike, Jacobi $(1 - r^2)^{|m|/2}$) for a later full-disk extension. Projected effort is five to six engineer-days on a dedicated branch; the validation plan reuses the Lane–Emden $n = 3$ benchmark, adds a Bessel-function analytic comparison for the Boussinesq annulus g-modes, and performs an approximate cross-check against GYRE’s $\ell = 1$ spectrum through the $\ell \rightarrow m$ correspondence.

8. Discussion and conclusions

Kiriko demonstrates that the uniform-density pseudospectral machinery extends cleanly to stratified stellar-interior flow provided three conditions are met. First, the pressure Poisson operator must be diagonalised in a basis adapted to the background density, which the Liouville-regularised reduced-pressure SL expansion accomplishes with a seven-step GPU pipeline that attains machine precision on the Boussinesq limit and the same relative accuracy at $N = 96$ as GYRE’s production sixth-order collocation scheme. Second, the interaction between the polytropic index and the Chebyshev basis must be acknowledged: only polynomial surface densities admit spectral convergence, and the distinction between $\sigma \in \mathbb{Z}$ and $\sigma \notin \mathbb{Z}$ is sharp rather than gradual. The Eddington polytrope’s unique convergence behaviour is a coincidence between astrophysical convention and approximation theory worth explicit note. Third, the time-domain operator must be assembled from the same discrete matrices that define the spatial eigenproblem; a primitive composition of node-space pointwise multiplications and differentiation matrices is not merely less accurate but structurally defective, leaking 5.1×10^{-5} per step on Lane–Emden $n = 3/2$ regardless of resolution. The full-Galerkin remedy was identified, proven in a Python driver at 5.1×10^{-18} dev/step, and specified as a three- to four-day CUDA port.

Together, these three results certify the CUDA `anelastic_sl_solver` as production-ready for the linear anelastic g-mode problem and provide a blueprint for the polar-annular successor. Beyond the immediate engineering outcome, the methodological pattern—small Python prototypes that foreclose or validate specific multi-day CUDA investments—is the single most productive habit visible in the development record, and is recommended as standard practice for all future paths.

Acknowledgements

The GYRE reference data in Section 4.4 was produced on a collaborator’s workstation running Fedora 43 with MESA SDK 26.3.2 and GYRE built from source; the SHA256 checksums of the reference artefacts are recorded in `gmode_exp_k_cuda_benchmark_2026-05-03.md` §5. All computations were performed on a single NVIDIA GeForce RTX 4080 SUPER (compute capability 8.9) with CUDA 12.9.86 and GCC 15.2.1. Source code and reproduction scripts are archived in the `anelastic-sl-spectral` branch of the `stellar2d` repository.

Primary source records

1. `docs/pseudo_spectral_design_2026-05-01.md` — uniform-density baseline
2. `docs/anelastic_SL_spectral_design.md` — initial SL design
3. `docs/anelastic_sl_phase0_2026-05-02.md` — Phase 0 viability study
4. `docs/reduced_pressure_liouville.md` — reduced-pressure derivation
5. `docs/anelastic_sl_cuda_phase1ab.md` — CUDA basis precompute and Poisson pipeline
6. `docs/polytropic_index_spectral_convergence_2026-05-03.md` — convergence cliff
7. `docs/gmode_experiments_2026-05-02.md` — experiments H, I, J, K
8. `docs/gmode_exp_k_cuda_benchmark_2026-05-03.md` — GYRE cross-validation
9. `docs/qspace_sl_path_b_2026-05-03.md` — Path B results
10. `docs/path_c_python_benchmark_2026-05-03.md` — Path C rejection
11. `docs/full_galerkin_closure_proof_2026-05-03.md` — Path D proof
12. `docs/phase2_operator_mismatch_summary_2026-05-03.md` — consolidated diagnosis
13. `docs/polar_anelastic_design_2026-05-03.md` — polar extension design